

GCP Resource Tagging: Enterprise Metadata Governance Platform

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1. Executive Summary

RB Global's GCP metadata governance solution is implemented as a **centralised, event-driven and serverless platform** for governing metadata across the Google Cloud estate. It supports existing resources (Brownfield) and newly created resources (Greenfield), while application ownership and required governance values remain outside runtime code.

Two dedicated governance projects provide environment isolation: **platform-metadata-dev** for development, integration and controlled validation, and **platform-metadata-prod** for production organisation-wide governance.

The design combines Cloud Asset Inventory, Cloud Audit Logs, Pub/Sub/Eventarc, Cloud Run, Cloud Tasks, Cloud Storage and BigQuery. This provides a consumption-based control plane, near-real-time Greenfield processing and auditable Brownfield remediation.

Core Platform Capabilities

Capability	Current Project Implementation
Brownfield Governance	Cloud Asset Inventory discovery -> compliance -> planning -> Cloud Tasks -> worker -> resource adapter
Greenfield Governance	Cloud Audit Logs -> Pub/Sub/Eventarc -> Cloud Run -> classifier -> compliance -> resource adapter
Application Registry	YAML source of truth in Cloud Storage with schema validation and runtime caching
Compliance Engine	Compares live metadata against registry-derived required metadata
Remediation	Service-specific adapters perform safe metadata updates using native GCP APIs
Audit & Reporting	BigQuery stores inventory, compliance, plans and execution evidence; dashboard consumes reporting APIs

2. Architecture and Processing Model

Brownfield and Greenfield use the same governance model. Their ingestion paths differ, but both converge on registry resolution, capability validation, compliance evaluation, service-specific remediation and audit evidence.

Mode	End-to-End Flow
Brownfield	Target projects -> Cloud Asset Inventory -> Discovery -> Registry -> Compliance -> Planner -> Cloud Tasks -> Worker -> Adapter -> BigQuery
Greenfield	Resource creation -> Cloud Audit Logs -> Pub/Sub/Eventarc -> Cloud Run -> Classifier -> Capability -> Compliance -> Adapter -> BigQuery

Serverless Design Benefits

Benefit	Outcome
Cost Efficiency	Cloud Run and Cloud Tasks scale with actual work; no dedicated always-on remediation server is required.
Near Real-Time Compliance	Greenfield creation events are evaluated after the corresponding audit event is delivered.
Zero Idle Worker Infrastructure	Asynchronous workers execute only when remediation batches exist.
Resilience	Independent batches, retries and failure isolation prevent one resource failure from stopping unrelated work.
Central Visibility	BigQuery and the dashboard provide a consolidated governance view across target projects.

3. Brownfield Governance

Brownfield processing governs existing resources across the selected project scope. Each execution creates a common **run_id**, discovers resources, evaluates supported capabilities, plans required changes and submits remediation work asynchronously.

The default **REMEDIATION_BATCH_SIZE** is **500**. Therefore, 1,000,000 planned remediation actions represent approximately 2,000 Cloud Tasks batches before retries - not one million individual tasks.

Brownfield Control Points

Control	Behaviour
Discovery	Cloud Asset Inventory provides scalable inventory discovery across visible target projects.
Registry Resolution	Project binding resolves the application and required governance metadata.
Capability Gate	Unsupported or disabled asset types are excluded from mutation.
Compliance	Actual metadata is compared with required values.
Planning	Only non-compliant resources requiring action become remediation plans.
Execution	Cloud Tasks delivers remediation batches to the authenticated worker endpoint.
Evidence	Execution result, status and mode are persisted for audit and reporting.

Runtime Controls

Setting	Default	Purpose
REMEDIATION_BATCH_SIZE	500	Groups remediation actions into enterprise-scale task batches.
MAX_PARALLEL_WORKERS	10	Controls application-side parallel execution.
PRESERVE_EXISTING_LABELS	true	Protects unrelated existing labels during reconciliation.
DRY_RUN	false	Can be enabled during controlled onboarding and validation.
EXCLUDED_PROJECTS	Configured	Prevents selected projects from governance processing.
EXCLUDED_BUCKETS	Configured	Protects registry/platform buckets from unintended remediation.

4. Greenfield Governance

Greenfield governance is event-driven. Resource creation activity is captured from Cloud Audit Logs and routed through the organisation-level event path into the governance service. The event is normalised by a classifier before the shared capability, compliance and remediation layers are invoked.

Adapter code existing in the repository does not by itself mean that a resource is Greenfield-supported. A resource is enabled only after its exact creation event, routing, classifier, capability entry, adapter behaviour, IAM and retry/idempotency characteristics are validated.

Greenfield Gate	Required Validation
Audit Event	Exact serviceName and methodName for creation are confirmed.
Organisation Routing	Events from governed workload projects reach the central event transport path.
Classifier	Event resolves to the correct canonical resource and asset type.
Capability	Asset type is explicitly enabled in supported resource configuration.
Adapter	Live metadata read/update behaviour is tested for the target service.
IAM	Runtime service account has only required read/update permissions.
Idempotency	Duplicate event delivery is safe and does not create destructive changes.

5. Registry, Data and Reporting

The **Application Registry** is the metadata source of truth. Application ownership and governance values are stored as YAML in Cloud Storage and validated against the registry schema before promotion. Runtime caching reduces repeated registry reads.

Typical attributes include product, team, owner, budget owner, organisation, department and cost centre. GCP bindings identify project, region, environment and business criticality.

BigQuery Governance Data

Object	Purpose
resource_snapshot	Discovered inventory and resource metadata snapshot.
compliance_snapshot	Point-in-time compliance evaluation evidence.
remediation_plan	Planned changes generated for a governance run.
remediation_execution	Execution status, evidence and BROWNFIELD/GREENFIELD execution mode.

Dashboard and Reporting

View	Purpose
Executive Summary	Estate size, supported resources, compliance and overall governance position.
Brownfield Summary	Planned, completed, remaining and failed remediation.
Greenfield Summary	Events, successful/failed processing and recent activity.
Project View	Compliance breakdown by governed project.
Resource Type View	Compliance by supported GCP resource family.
Recent Runs	Operational execution history.
Top Non-Compliant Resources	Prioritised resources with missing or incorrect governance metadata.

6. Resource Capability Model

Runtime enablement is centrally controlled by the supported-resource configuration. This separation matters because the repository can contain clients or adapters that are under development, disabled or unsuitable for labels.

The repository contains integration code for multiple GCP service families, including Compute Engine, BigQuery, Cloud Storage, Resource Manager, Cloud KMS, Pub/Sub, GKE, Cloud Run, Cloud SQL, Artifact Registry, Cloud Functions, Vertex AI, Dataplex, App Engine, Redis, Secret Manager, Monitoring, Dataform, DNS, AlloyDB and API Keys.

Rule	Requirement
Code Presence != Support	An adapter/client in the repository is not sufficient evidence of production support.
Explicit Enablement	The asset type must be present in the active label/tag capability set.
Test Before Promotion	Brownfield and, where applicable, Greenfield behaviour must be validated in platform-metadata-dev.
IAM Follows Capability	Permissions are granted for enabled resource types rather than every candidate integration.
Tags Separate from Labels	Resource Manager Tag capability is governed independently from service label capability.

7. Dedicated Project and Environment Model

Area	platform-metadata-dev	platform-metadata-prod
Purpose	Development, integration, adapter/event testing and controlled scale validation	Production organisation-wide metadata governance
Runtime Identity	Dedicated DEV service account	Dedicated PROD service account
Registry	DEV registry/configuration	Production-approved registry/configuration
BigQuery	DEV governance dataset	Production governance/audit dataset
Event / Queue	DEV-specific	PROD-specific
Access Boundary	Must not unintentionally remediate PROD workloads	Independent production trust boundary

8. IAM and Security Model

The governance runtime uses a dedicated service account and follows least privilege. The final client custom role should be generated from the **resource capabilities actually enabled for production**, not simply from every API client present in the repository.

Category	Required Access
Organisation / Project Discovery	Read access required to enumerate authorised target projects and resolve project metadata.
Cloud Asset Inventory	Read/search access required for Brownfield resource discovery.
Registry	Read access to approved Cloud Storage registry objects.
BigQuery	Dataset/table access required for governance evidence and reporting.
Cloud Tasks	Permission to enqueue remediation batches and authenticated invocation for the worker path.
Resource Metadata	Service-specific read/update permissions only for enabled adapters.
Resource Manager Tags	Tag permissions only when Tag governance is explicitly enabled.
Dashboard	Restricted to authorised organisation users; operational worker endpoints remain service-to-service.

9. Approach Comparison

Criteria	Earlier Native-Only Business Case	Current Enterprise Platform
Runtime	No application runtime	Modular Cloud Run governance engine
Brownfield Detection	CAI export + BigQuery SQL	CAI discovery + live metadata enrichment + compliance engine
Brownfield Remediation	Terraform/gcloud operational workflow	Planner + Cloud Tasks + worker + service-specific adapters
Greenfield	Terraform + Organisation Policy prevention	Near-real-time audit-event-driven detection and remediation
Registry	Application Registry used as value source	Validated YAML source of truth loaded dynamically from Cloud Storage
Reporting	BigQuery + Looker Studio concept	BigQuery-backed reporting APIs and integrated dashboard
Scale Model	Operator/script based	Asynchronous serverless batch processing

10. Recommendation and Production Readiness

Proceed with the **Enterprise Metadata Governance Platform**, using **platform-metadata-dev** as the mandatory validation environment and **platform-metadata-prod** as the production control plane.

Production onboarding should be capability-by-capability. Each resource family must have verified discovery, metadata behaviour, least-privilege IAM, retry/idempotency and - for Greenfield - exact creation-event routing before enablement.

Production Gate	Required Outcome
Registry	Schema validation and unique project bindings pass before promotion.
Capability	Only tested resource types are enabled.
IAM	Least-privilege permissions match enabled capabilities.
Brownfield	Batching, retry, failure isolation and representative scale are validated.
Greenfield	Exact creation events and end-to-end organisation routing are validated.
Security	Dashboard access and internal service endpoints are authenticated and restricted.
Operations	Logging, alerting, budgets, rollback and runbook are approved.

Important - Capability Control: Resource support is determined by active capability configuration. The presence of an adapter, client or IAM permission must never be treated as proof that a resource is production-enabled.